

Appendix S2. Preregistered experimental design and analysis plan

ENA-ABM v0.5 main theoretical experiment

This appendix documents the design frozen on 25 July 2026 before execution and substantive interpretation of the version 0.5 computational experiment. The study uses the version 0.4 agent-based model without adding behavioral mechanisms. Its purpose is theoretical: all profiles and parameter ranges are diagnostic design inputs, not empirically calibrated estimates or observations sampled from a population. Appendix S1 describes the model; Appendix S3 provides the complete machine-readable parameter and configuration dictionary.

S2.1 Research question and model boundary

The research question is:

Under what combinations of market sociality, network structure, consumer conditions, entrepreneurial resources, and referral opportunities does one-to-one referral activation equal or outperform direct outreach under an equal allocated budget?

The comparison includes only two strategies: direct outreach and one-to-one referral outreach. The frozen model excludes promoters, mass referrals, events, negative word of mouth, combined strategies, incentives as behavioral mechanisms, Consumat decision logics, competition, and empirical calibration. Valid low-adoption, zero-adoption, extinction, unavailable-opportunity, and resource-constrained runs are retained because they represent theoretical boundary conditions rather than simulation failures.

S2.2 Estimands and outcomes

For parameter profile x and referral cost ratio $r = C_R / C_D$, let $Y_R(r, x)$ and $Y_D(x)$ denote the numbers of new adopters generated by referral and direct outreach, excluding seed adopters. The expected paired difference is

$$\Delta(r \mid x) = E[Y_R(r, x) - Y_D(x)] \quad (\text{S2.1})$$

The primary profile-specific competitive threshold is

$$r^*(x) = \sup \{ r : \Delta(r \mid x) \geq 0 \} \quad (\text{S2.2})$$

Operationally, r^* is the largest evaluated cost ratio with a nonnegative mean paired adoption difference, provided that the evaluated curve yields an interpretable competitive region.

Four threshold definitions were fixed before the main analysis:

Estimand	Operational definition
Primary adoption threshold	Largest evaluated r for which the mean paired difference $\bar{\Delta}(r \mid x)$ is nonnegative.
Conservative threshold	Largest evaluated r for which the lower limit of the 95% interval for the paired adoption difference is nonnegative.
Probability threshold	Largest evaluated r for which $\Pr(Y_R > Y_D) \geq 0.50$.
Allocated-budget efficiency threshold	Largest evaluated r for which the difference in new adoptions per allocated budget unit is nonnegative. Because both strategies receive the same allocated budget, this threshold is algebraically aligned with the primary adoption threshold.

The primary outcome is new adopters excluding seeds. Predeclared secondary outcomes include attempts, unavailable opportunities, accepted referral requests, eligible neighbors, effective introductions, effective

exposures, adoption per exposure, budget remaining, takeoff, extinction, time to first nonseed adoption, mean adoption time, referral-cascade reach, maximum cascade depth, and largest referral cascade.

S2.3 Main experimental design

S2.3.1 Profile construction

The main design contains 96 deterministic stratified Latin-hypercube profiles. This space-filling design covers the full screened ranges of nine prioritized factors without requiring a large full factorial. The three network structures are balanced at 32 profiles each: ring lattice, Watts–Strogatz small world, and preferential attachment.

Each continuous factor is stratified across its complete range, and the deterministic design assigns one value from each stratum. Integer-valued budget and seed-adopter settings are mapped to valid integer values. The resulting profile registry is fixed and identified as P001–P096. The complete row-level registry is supplied separately as `registered_profiles_v0.5.csv` rather than reproduced in this narrative appendix.

S2.3.2 Varied factors

Factor	Configuration field	Range
Market sociality	market_sociality	0.00–1.00
Mean product fit	fit_mean	0.35–0.80
Mean status-quo satisfaction	status_mean	0.20–0.80
Mean initial uncertainty	uncertainty_mean	0.20–0.85
Adoption intercept	beta_0	–1.50–0.50
Individual-fit coefficient	beta_individual	1.00–6.00
Social-influence coefficient	beta_social	0.00–6.00
Allocated budget	budget_limit	8–60 direct-cost units
Initial seed adopters	seed_adopters	1–8

S2.3.3 Fixed settings in the main profiles

Component	Main-study setting
Consumers	100
Mean degree	6
Small-world rewiring probability	0.10
Horizon	30 ticks
Entrepreneurial capacity	2 actions per tick
Product-fit standard deviation	0.15
Minimum uncertainty	0.05
Information-update coefficient	0.25
Direct and referral credibility	0.60 / 0.60
Direct effective-exposure probability	0.512
Referral acceptance probability	0.80
Referral eligible-neighbor availability	0.80
Referral effective-introduction probability	0.80
Primary takeoff share	0.10
Primary extinction quiet period	5 ticks

The neutral access benchmark equates the direct effective-exposure probability, 0.512, with the product of the three conditional referral probabilities, $0.80^3 = 0.512$. This controls conditional exposure probability without eliminating structural differences in access.

S2.4 Relative-cost design and paired simulation

Direct-attempt cost is normalized to $C_D = 1$, while referral-attempt cost is $C_R = r$. The cost ratio affects only resource consumption; it does not alter acceptance, exposure, credibility, utility, or adoption probability.

The primary accounting convention charges the full expected action cost when a feasible attempt begins. A failure after initiation remains costly. A referral action that cannot begin because no adopted source exists is recorded as an unavailable opportunity and consumes no cost. Staged accounting is reserved for robustness analysis.

The coarse cost grid is

$$G = \{0.25, 0.50, 0.75, \dots, 2.75, 3.00\} \quad (\text{S2.3})$$

Every adjacent coarse-grid interval containing a sign change is evaluated at intermediate increments of 0.05. The algorithm searches every bracket rather than assuming monotonicity or interpolating a smooth response.

Each profile-cost cell uses 400 paired replications. Within a profile and replication, direct and referral runs share the network realization, consumer attributes, seed adopters, initialization namespace, and relevant random streams. The same profile-replication initialization is also aligned across evaluated values of r . Pairing makes the primary observation

$$d_{k,r}(x) = Y_{R,k}(r, x) - Y_{D,k}(x) \quad (\text{S2.4})$$

for paired replication k , thereby reducing variation caused by comparing unrelated simulated markets. The main experiment seed is 2026072505.

S2.5 Threshold-estimation algorithm and classification

Stage	Prespecified operation
Input	Profile x ; coarse grid G ; 400 paired replications; refinement increment 0.05; 500 paired bootstrap samples.
1	Simulate direct and referral strategies for every coarse-grid value using matched initialization and random streams; store $d_{k,r}(x)$.
2	Calculate $\bar{D}(r \mid x)$ across the 400 paired replications.
3	Inspect every adjacent pair of coarse-grid cells and retain every sign-change bracket.
4	Evaluate all 0.05-spaced intermediate values within every retained bracket using the same paired design.
5	Classify the profile and, when appropriate, set r^* to the largest evaluated r with nonnegative mean paired difference.
6	Resample paired differences 500 times within the profile and re-estimate the threshold to quantify Monte Carlo uncertainty.

Profiles are classified as follows:

- **Finite threshold:** the evaluated curve contains an interpretable competitive region and a finite upper boundary.
- **No competitive region:** no evaluated value of r produces a nonnegative mean paired difference.
- **Right-censored:** referrals remain competitive through the upper tested boundary, $r = 3.00$.
- **Multiple crossing:** the curve enters and leaves the competitive region more than once; all crossings are retained and a single threshold is not forced.

The classification preserves absent and censored regions as scientific outcomes rather than replacing them with interpolated values.

S2.6 Confirmatory summaries

The following summaries were declared before substantive interpretation:

1. the distribution, median, interquartile range, and finite range of $r^*(x)$;
2. the share of registered profiles in which referrals are competitive at $r = 1.00$;
3. the share competitive at the version 0.4 reference value $r = 1.20$;
4. mean paired adoption differences and 95% intervals at $r = 1.00$ and $r = 1.20$;
5. differences in takeoff, extinction, exposure, conversion, cascade, and timing measures;
6. threshold distributions stratified by network structure; and
7. conditions associated with finite, absent, right-censored, or multiple-crossing competitive regions.

The distribution of profile thresholds is not used to infer one universal r^* or the prevalence of referral advantage in real markets.

S2.7 Uncertainty and sensitivity plan

Monte Carlo uncertainty is separated from variation across parameter profiles:

- **Paired bootstrap:** 500 within-profile resamples of the 400 paired differences quantify threshold stability.
- **Spearman associations:** rank correlations describe monotonic associations between the nine design factors and finite thresholds.
- **Nonlinear emulator:** a deterministic random-forest surrogate summarizes nonlinear variation using permutation importance and cross-validated predictive accuracy.
- **Descriptive classification:** shallow decision trees summarize combinations associated with referral competitiveness at $r = 1.00$ and $r = 1.20$.

Permutation importance and decision-tree rules are treated as exploratory descriptions, not causal effects. A continuous predictive surface is warranted only if out-of-sample emulator performance is adequate.

S2.8 Preregistered robustness analyses

S2.8.1 Exposure-process robustness

Eight regimes are evaluated separately in each of the three network structures.

Regime	Direct exposure	Referral acceptance	Eligible-neighbor availability	Effective introduction
Neutral	0.512	0.80	0.80	0.80
Equal high	0.729	0.90	0.90	0.90
Direct-exposure advantage	0.80	0.80	0.80	0.80
Referral-exposure advantage	0.512	0.90	0.90	0.90
Low acceptance	0.512	0.40	0.80	0.80
High acceptance	0.512	1.00	0.80	0.80
No eligible neighbors	0.512	0.80	0.00	0.80
Abundant eligible neighbors	0.512	0.80	1.00	0.80

S2.8.2 Accounting robustness

Full expected cost at initiation is compared with staged accounting in nine predefined contexts: baseline, low sociality (0.10), high sociality (0.90), scarce budget (8), abundant budget (60), scarce referrers (one seed), abundant referrers (eight seeds), ring network, and preferential-attachment network. The staged convention uses four equal diagnostic activity shares of 0.25. These shares are theoretical and are not presented as empirical cost estimates.

S2.8.3 Structural robustness

The structural analysis evaluates baseline conditions and eight variants: 50 and 200 consumers; mean degree 4 and 10; rewiring probabilities 0.00 and 0.30; ring network; and preferential-attachment network. Only existing structural parameters are changed; no behavioral mechanism is added.

S2.8.4 Dynamic-definition robustness

Takeoff and extinction definitions are evaluated at $r = 1.00$ and $r = 1.20$. Takeoff shares are 5%, 10%, 15%, and 20%; extinction quiet periods are 3, 5, and 10 ticks. These variants change classification of dynamic outcomes but do not alter simulated exposure or adoption events.

S2.9 Interpretation stop rules

Condition	Prespecified stopping criterion
Multiple crossings	Stop substantive threshold interpretation if more than 10% of main profiles exhibit multiple crossings.
Interpretable regions	Stop if fewer than 50% of profiles yield an interpretable competitive region.
Bootstrap stability	Stop if bootstrap threshold width exceeds 0.50 for a majority of finite profiles.
Budget residuals or rounding	Stop if residual effects exceed 10% and determine threshold ordering.
Accounting convention	Stop if upfront and staged thresholds differ by at least 0.25 across enough central contexts to reverse the substantive conclusion.
Mechanically dominant process	Stop if a process probability determines the comparison so strongly that no conditional interpretation remains.
Reproduction	Stop if an independent complete reproduction does not match.

An individual profile or boundary condition may legitimately lack a competitive region without stopping the study. Any unplanned design change must be documented before the affected analysis is executed or interpreted.

S2.10 Documented protocol correction

The frozen design declared a confirmatory comparison at $r = 1.20$, inherited from the version 0.4 reference threshold. The configured coarse grid contained 1.00 and 1.25 but not 1.20. Local refinement produced 1.20 only for profiles whose crossing bracket included that value, so the registered contrast could not initially be calculated for all profiles.

The inconsistency was documented on 25 July 2026 after completion of the main cost grid but before sensitivity analysis, competitiveness classification, robustness analysis, or substantive interpretation. One additional paired cell at $r = 1.20$ was therefore executed for every registered profile using the unchanged model, the same experiment seed and profile-specific initialization namespace, 400 paired replications, and upfront expected-cost accounting.

The added comparison was stored separately as `main_fixed_cost_1.20_runs_v0.5.csv.gz` and `main_fixed_cost_1.20_summary_v0.5.csv`. It did not alter the coarse grid, crossing refinement, primary

thresholds, bootstrap thresholds, parameter ranges, replication count, stopping rules, or profile selection. The correction completed an already declared contrast and was not motivated by observed strategy performance.

S2.11 Design artifacts and appendix boundaries

Artifact	Role in the supplementary package
PREREGISTERED_DESIGN_v0.5.md	Frozen narrative design and interpretation rules; principal source for Appendix S2.
main_study_v0.5.json	Machine-readable design, robustness contexts, seeds, and stopping-rule values; belongs with Appendix S3.
registered_profiles_v0.5.csv	Complete P001–P096 profile registry; belongs with Appendix S3.
PROTOCOL_CORRECTION_v0.5.md	Timestamped explanation and scope of the $r = 1.20$ correction; cross-referenced in S2.10.
DECISION_LOG_v0.5.md	Rationale for frozen choices and subsequent execution decisions.
Reproducible package and hashes	Software, commands, and verification materials; documented in Appendices S6–S7.

Appendix S2 records the design and analysis plan. It intentionally excludes the observed threshold distribution, robustness findings, sensitivity rankings, and reproduction results, which are reported in the article and subsequent supplementary appendices.